



What is Threat Modeling?

- A structured approach to identify, analyze, and mitigate security threats in a system
- Helps security teams anticipate attacks before they happen
- Focuses on thinking like an attacker during system design
- Core goal: build secure systems proactively

Why Threat Modeling Matters

- Identifies security risks early in the development lifecycle
- Reduces cost of fixing vulnerabilities later
- Improves secure architecture and design decisions
- Helps organizations prioritize high-impact threats



Security Mindset for Threat Modeling

- Always ask: “What could go wrong?”
- Assume attackers will abuse every exposed component
- Focus on attack surfaces and trust boundaries
- Security should be built-in, not bolted-on



Core Components of Threat Modeling

- **Assets** → What needs protection (data, systems, credentials)
- **Threats** → Potential attacks against assets
- **Vulnerabilities** → Weaknesses attackers exploit
- **Attack Vectors** → Paths attackers use to reach targets
- **Mitigations** → Security controls to reduce risk



The STRIDE Threat Model

- Common framework used in threat modeling:
- **S – Spoofing** → Impersonating another user/system
- **T – Tampering** → Unauthorized data modification
- **R – Repudiation** → Denying performed actions
- **I – Information Disclosure** → Sensitive data leakage
- **D – Denial of Service** → System availability disruption
- **E – Elevation of Privilege** → Gaining higher permissions



Data Flow Diagrams

- Used to visualize system architecture.
- Key elements:
 - Processes → System operations
 - Data Stores → Databases or storage
 - External Entities → Users or external systems
 - Data Flows → Information movement
- Helps identify security risks across system components.

Trust Boundaries

- Points where data moves between different trust levels
- Common attack points in systems
- Internet → Internal network
- User input → Application server
- API → Database
- Security controls must be applied at these boundaries.

Threat Modeling Process

- Define system architecture
- Identify assets and entry points
- Map data flows and trust boundaries
- Identify threats using frameworks
(like STRIDE)
- Apply security mitigations

Common Threat Modeling Techniques

- STRIDE → threat categorization
- PASTA → risk-driven threat modeling
- Attack Trees → visualize attack paths
- Kill Chain Mapping → attacker lifecycle analysis

Real-World Use Cases



Interview-Level Key Concepts



Common Mistakes in Threat Modeling

Key Takeaway